



Report on Advanced SOC Operations, Threat Intelligence Integration, and Incident Response

- **Advanced Log Analysis**

- Advanced log analysis is one of the most important responsibilities in a Security Operations Center (SOC). Every device, application, operating system, firewall, and security solution generates logs that record activities occurring inside the environment. These logs contain valuable information about user behavior, system activity, authentication attempts, network communication, application execution, and possible security incidents.
- SOC analysts examine these logs to detect suspicious activities, identify attacks, investigate incidents, and understand attacker behavior. Basic log analysis may involve reviewing a single log source, but advanced log analysis focuses on correlating events from multiple systems to uncover complex attack patterns that are difficult to detect individually.

1. Log Correlation

- Log correlation is the process of combining logs from multiple systems and identifying relationships between events that occur across different devices or applications. Attackers rarely interact with only one system during an attack. Their activities often generate traces across endpoints, firewalls, DNS servers, VPNs, authentication systems, and cloud services.
 - When logs are analyzed separately, suspicious activity may appear harmless. However, when events are connected together chronologically and contextually, analysts can identify full attack chains.
- Log correlation allows analysts to
 - connect related events
 - reconstruct attack timelines
 - identify lateral movement
 - detect coordinated attacks
 - improve investigation accuracy
 - automate alert generation

2. Anomaly Detection



- Anomaly detection focuses on identifying behavior that deviates from normal system activity. Instead of searching for known attack signatures only, anomaly detection identifies unusual patterns that may indicate malicious activity.
- Attackers often try to hide inside legitimate traffic and user behavior. Anomaly detection helps SOC analysts identify subtle threats that signature-based tools may miss.

3. Log Enrichment

- Raw logs often contain limited information. Log enrichment enhances logs by adding additional context that helps analysts understand the severity and meaning of events more effectively.
 - Enriched logs provide more actionable intelligence during investigations.
- Purpose of Log Enrichment
 - improve alert accuracy
 - reduce investigation time
 - provide business context
 - prioritize incidents better
 - improve threat visibility

1. Geolocation Enrichment

- One common enrichment technique is IP geolocation.
- If an organization only operates in India, foreign logins may require investigation.

2. Threat Intelligence Enrichment

Threat intelligence feeds provide information about:

- malicious IP addresses
- phishing domains
- ransomware infrastructure
- malware hashes
- botnet activity

3. User Role Enrichment

- Adding user role information helps analysts understand business impact.
- Administrative accounts are more attractive to attackers because they provide elevated privileges.

4. Asset Criticality Enrichment



- Systems are prioritized based on importance.
- This helps SOC analysts prioritize incidents more effectively.

- **Threat Intelligence Integration**

- Threat intelligence integration is a critical component of modern Security Operations Centers (SOC). Organizations face constantly evolving cyber threats from ransomware groups, phishing campaigns, nation-state attackers, insider threats, and advanced persistent threats (APTs). Threat intelligence helps SOC analysts understand attacker behavior, identify malicious infrastructure, and improve detection capabilities.

1. Threat Intelligence Types

Threat intelligence is categorized into several types depending on the information provided and its purpose.

- **Indicators of Compromise (IOCs)**

- Indicators of Compromise (IOCs) are observable pieces of evidence that indicate malicious activity or system compromise.
- IOCs help SOC analysts detect known threats quickly.
- Ioc includes : malicious IP addresses, suspicious domains, malware hashes, phishing URLs, file names, registry changes, email addresses, network signatures.

- **Tactics, Techniques, and Procedures (TTPs)**

- TTPs describe how attackers behave during cyberattacks.
- Unlike IOCs, which may change frequently, attacker techniques and behaviors often remain consistent.
- It helps to understand : understand attacker methodology, identify attack stages, perform, threat hunting ,improve detection engineering

- **Threat Feeds**

- Threat feeds provide continuously updated intelligence data containing malicious Ips, domains, malware hashes, ransomware indicators, phishing infrastructure.
- Threat feeds are integrated into SIEM and SOC platforms for automated detection.
- Common Threat Intelligence Sources : AlienVault OTX, MITRE Corporation ATT&CK ,CISA advisories, VirusTotal, AbuseIPDB, MISP



2. Integration in SOC

Threat intelligence integration improves: threat detection alert accuracy, incident response, SOC visibility, investigation speed

Integrated intelligence helps analysts identify malicious activity automatically.

- Step-by-Step Workflow
 - The automation bridges the gap between raw data and actionable intelligence through four main steps:

Step 1: Threat Intel Ingestion

- The SOC hooks up the SIEM to global threat intelligence feeds. These feeds contain updated lists of IoCs (Indicators of Compromise) like malicious IPs, URLs, domain names, and file hashes. These can come from open-source platforms (like MISP, Abuse.ch) or commercial vendors (like CrowdStrike, Recorded Future), typically delivered automatically via STIX/TAXII protocols.

Step 2: Live Log Collection

- Simultaneously, your organization's infrastructure (firewalls, EDR agents, active directories) is sending millions of live event logs into the SIEM in real-time.

Step 3: Correlation & Matching

- The SIEM runs correlation rules. If a firewall log shows an internal laptop talking to an external IP, and that exact same IP exists on the downloaded "Bad/Malicious IP" threat intel list, a match is found instantly.

Step 4: Context-Rich Alerting

- Instead of a vague notification, the analyst receives an Enriched Alert that looks like this:
 - **Severity:** Critical - Connection to Known Ransomware C2 Server
 - **Source Internal Host:** 10.0.4.15 (Finance Desktop)
 - **Destination Malicious IP:** 198.51.100.42
 - **Threat Actor/Campaign:** APT29 (Cozy Bear)
 - **Confidence Score:** 98%



3. Threat Hunting with Intelligence

- Threat hunting is the proactive search for hidden threats that may bypass automated detection systems.
- Threat intelligence guides threat hunters by providing attacker TTPs and indicators.
- SOC analysts use intelligence to search for : suspicious behavior, attacker persistence, compromised accounts, lateral movement, malware execution.
- Example Scenario : A user account normally logs in from Pune during business hours.
 - Threat hunters investigate for possible credential compromise.
- Hunting for PowerShell Abuse : Threat hunters search endpoint logs for:
 - powershell.exe -EncodedCommand
- Hunting for Command-and-Control Traffic : Analysts search firewall and DNS logs .
For
 - suspicious domains
 - known C2 IPs
 - beaconing patterns
 - unusual outbound traffic
- **Incident Escalation Workflows**
 - Incident escalation workflows are critical processes within a Security Operations Center (SOC). Not every security alert can be handled by a single analyst or team. Some incidents are simple and low-risk, while others involve advanced attacks, critical business systems, or widespread compromise that require specialized expertise and management involvement.

1. Escalation Tiers

- Most Security Operations Centers use a tiered analyst model to organize incident handling responsibilities.
- Tier 1 – Alert Monitoring and Triage :



- Tier 1 analysts are the first responders in the SOC. They continuously monitor alerts generated by: SIEM platforms, firewalls, IDS/IPS systems, endpoint detection tools, email security systems
- Key Tasks Performed by Tier 1 Analysts : review incoming alerts, validate whether alerts are legitimate, identify false positives, collect initial evidence, classify incidents, determine severity levels, escalate suspicious incidents.
- Tier 2 – Incident Investigation :
 - Tier 2 analysts perform deeper investigations on escalated incidents. They analyze : attack behavior, malware activity, affected systems, attacker persistence, lateral movement.
 - Key Tasks Performed by Tier 2 Analysts : correlate logs, analyze endpoints, perform IOC searches, investigate malware execution, examine network traffic, validate attacker activity, recommend containment actions.
- Tier 3 – Advanced Analysis and Threat Hunting :
 - Tier 3 analysts are senior security specialists responsible for handling highly sophisticated threats.
 - They focus on : advanced persistent threats (APTs), malware reverse engineering complex forensic analysis, advanced threat hunting, detection engineering.
 - Key Tasks Performed by Tier 3 Analysts : analyze sophisticated malware, identify attacker TTPs, develop detection rules, conduct memory forensics, improve SOC detections, perform advanced investigations
- **Escalation Criteria**
 - Incidents are escalated when they exceed the capabilities or authority of the current analyst tier.
 - Escalation decisions depend on : severity, impact, complexity, attacker sophistication, affected assets, business criticality
- 2. **Communication Protocols**
 - Effective communication is critical during incident escalation. Security teams, management, legal departments, and executives may all require updates during major incidents.
 - Poor communication can : delay response, create confusion, increase business impact, cause inconsistent actions.

- **Situation Reports (SITREP)**



- SITREP stands for : Situation Report.
- A SITREP provides structured updates during incidents.
- A SITREP typically contains : incident summary, affected systems, severity level , investigation findings, containment status, current risks, recommended actions.
- **stakeholder briefings.**
 - Different stakeholders require different levels of information.
 - Technical Teams require : IOCs, logs, forensic evidence, attack details.
 - Management Teams require : business impact, operational risks, recovery estimates response progress.
 - Legal and Compliance Teams require : breach details, affected data, regulatory implications, evidence preservation status.

3. Automation in Escalation

- **SOAR Platforms**
 - Security Orchestration, Automation, and Response (SOAR) platforms automate escalation workflows.
 - Popular SOAR tools include : Splunk SOAR, Palo Alto Cortex XSOAR, IBM Resilient
 - SOAR automation helps organizations respond to threats more effectively. Proper escalation ensures that incidents are investigated by the appropriate experts, stakeholders remain informed, and security teams can minimize damage caused by cyberattacks.
- **Ticketing Systems** : SOC teams commonly use : Atlassian Jira, ServiceNow, TheHive. to track escalated incidents.
 - Many major breaches worsened because incidents were not escalated properly.
 - Efficient escalation workflows significantly improve incident response effectiveness.



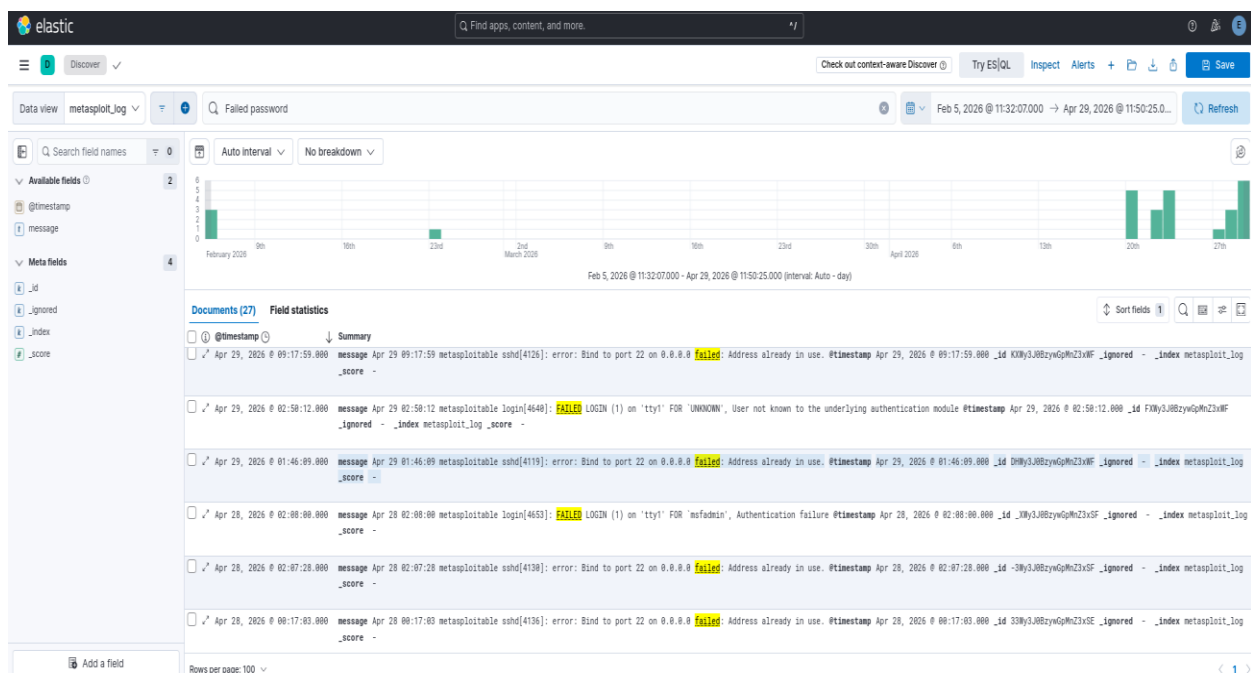
- **Advanced Log Analysis**

- The objective of this task was to understand advanced Security Operations Center (SOC) workflows by performing log correlation, anomaly detection, and log enrichment. The task focused on identifying suspicious activity through event correlation, detecting abnormal traffic behavior, and enriching logs with additional contextual information such as geolocation.

1. Log Correlation

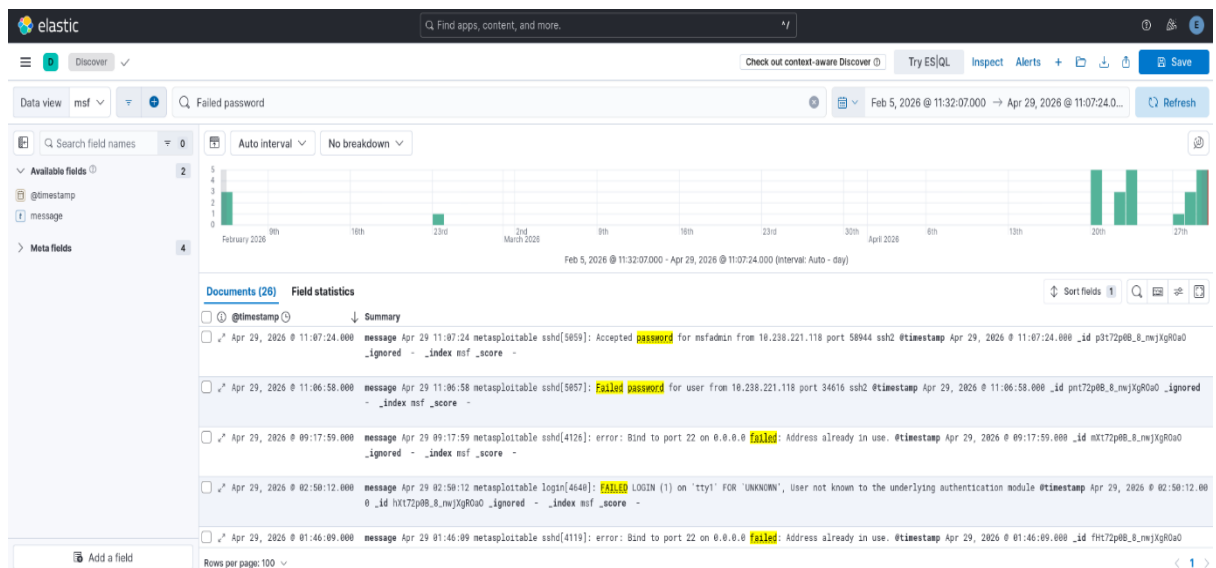
- The purpose of log correlation was to identify suspicious relationships between failed authentication events and outbound network communication.
- Repeated failed login attempts followed by external communication may indicate : Credential attacks, Malware beaconing, Data exfiltration Unauthorized access attempts

- **Log correlation in kibana**





- Failed password detection



- The analysis showed that failed login attempts were closely followed by outbound encrypted traffic. This behavior may indicate suspicious communication initiated after unauthorized access attempts.
- The repeated communication with external IP addresses highlighted the importance of correlating authentication logs with network traffic to detect potential threats.

2. Anomaly Detection

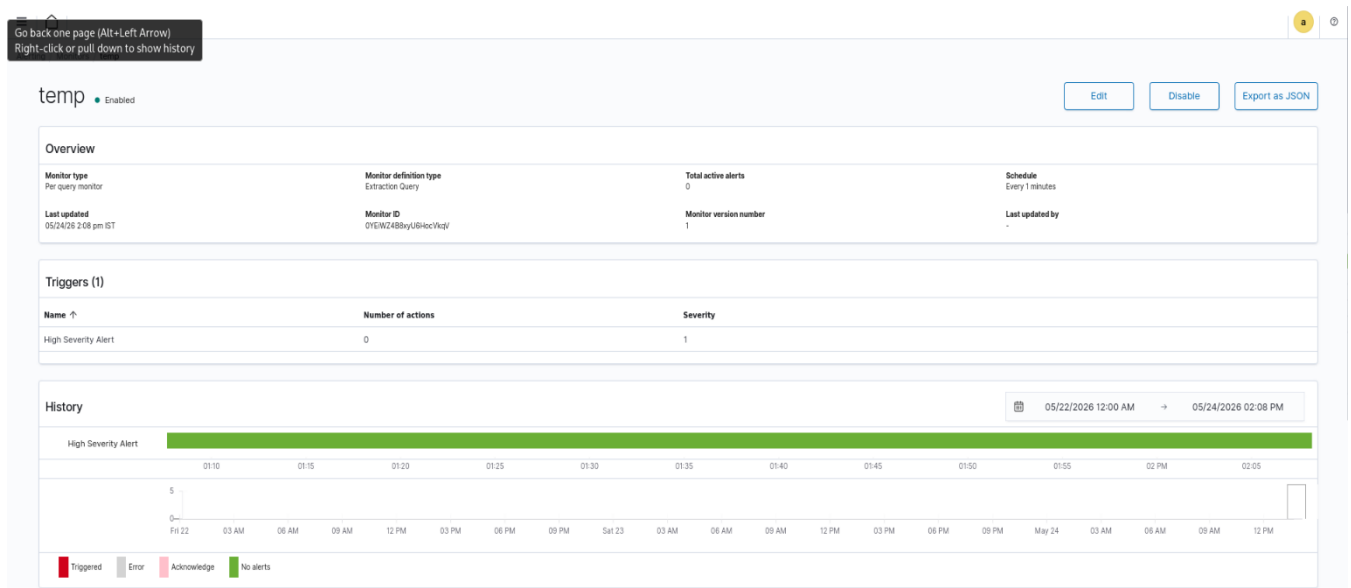
- The objective of anomaly detection was to identify unusual network behavior such as large outbound data transfers occurring within a short period of time.
- Modern cybersecurity monitoring relies heavily on behavioral analysis rather than only signature-based detection. SOC analysts use anomaly detection to identify suspicious activities such as unusually large data transfers, traffic spikes, or abnormal communication patterns.

- High-volume traffic was generated using

```
curl http://speedtest.tele2.net/10MB.zip -o testfile.zip
```



- rule created in Elasticsearch



- The generated traffic produced noticeable spikes in packet transmission and bandwidth usage. Such sudden increases in network activity are often treated as suspicious in SOC environments because attackers commonly transfer data rapidly after compromising systems.
- The exercise demonstrated how anomaly detection helps security analysts identify abnormal behaviors even when no known attack signature exists.

3. Log Enrichment

- The purpose of log enrichment was to improve the quality of network logs by adding geographical information to external IP addresses.
- GeoIP enrichment helps analysts : Identify the origin of traffic, Detect unusual foreign communication, Improve threat investigations, Add context to alerts.
- The GeoIP utility was installed using :

```
sudo apt install geoip-bin -y
```



- installing a plugin and adding a geolocation to IP

```
(root@kali)~# sudo apt install geoiplib-bin -y
The following packages were automatically installed and are no longer required:
  libwireshark18 libwireshark15 libwsutil16
Use 'sudo apt autoremove' to remove them.

Installing:
  geoiplib-bin

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1864
  Download size: 24.0 kB
  Space needed: 61.4 kB / 26.5 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 geoiplib-bin amd64 1.6.12-11.1+b2 [24.9 kB]
Fetched 24.9 kB in 9s (2,887 B/s)
Selecting previously unselected package geoiplib-bin.
(Reading database ... 739817 files and directories currently installed.)
Preparing to unpack .../geoiplib-bin_1.6.12-11.1+b2_amd64.deb ...
Unpacking geoiplib-bin (1.6.12-11.1+b2) ...
Setting up geoiplib-bin (1.6.12-11.1+b2) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for man-db (2.13.1-1) ...

(root@kali)~# geoipliblookup 52.204.151.28
GeoIP Country Edition: US, United States

(root@kali)~# geoipliblookup 104.208.16.92
geoipliblookup 18.211.27.106
GeoIP Country Edition: US, United States
GeoIP Country Edition: US, United States
```

- GeoIP enrichment provided valuable context for analyzing external network communication. Instead of viewing only raw IP addresses, the analysis revealed the geographical origin of the connections.
- This improved visibility and helped distinguish potentially suspicious communication from expected network activity.

• Threat Intelligence Integration

- The objective of this task was to understand how Security Operations Centers (SOCs) integrate external threat intelligence feeds into security monitoring platforms to detect malicious activity. The task also focused on alert enrichment and threat hunting techniques using simulated Wazuh and AlienVault OTX integration.
- Threat intelligence integration improves detection capabilities by correlating internal events with external intelligence sources. Additionally, alert enrichment adds contextual information to alerts, helping analysts prioritize investigations and improve incident response efficiency.

1. Threat Feed Integration

- The goal of this section was to simulate the integration of threat intelligence feeds into Wazuh for detecting known malicious IP addresses.



- Custom Wazuh Detection Rule

```
root@kali: /home/kali
GNU nano 8.7 /var/ossec/etc/rules/local_rules.xml
group name="custom_ssh_bruteforce,">

<rule id="100100" level="10" frequency="3" timeframe="120">
  <if_matched_sid>5710</if_matched_sid>
  <same_source_ip />
  <description>Multiple failed SSH login attempts detected</description>

  <mitre>
    <id>T1110</id>
  </mitre>
</rule>
</group>

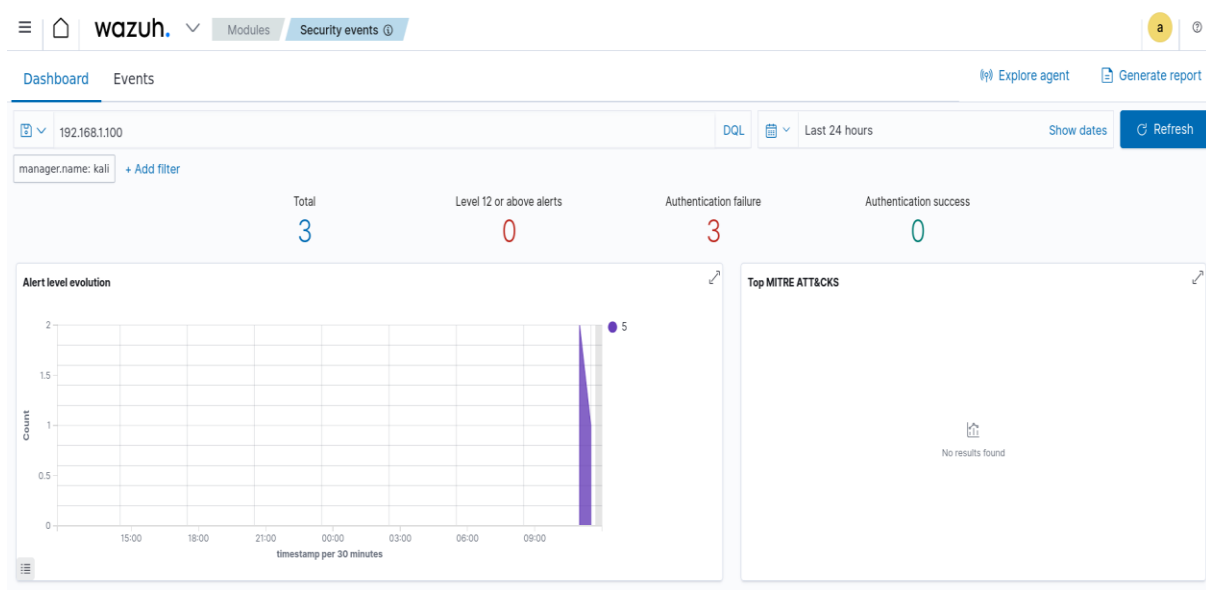
<group name="threat_intel,">
  <rule id="100200" level="12">
    <decoded_as>sshd</decoded_as>

    <list field="srcip" lookup="match_key">
      etc/lists/alienvault_reputation.ipset
    </list>

    <description>
      AlienVault OTX: Malicious IP detected
    </description>

    <mitre>
      <id>T1110</id>
    </mitre>
  </rule>
</group>
```

- Wazuh result





2. Alert Enrichment

- The purpose of alert enrichment was to improve security alerts by adding contextual threat intelligence information.

- **Enriched Alert Table**

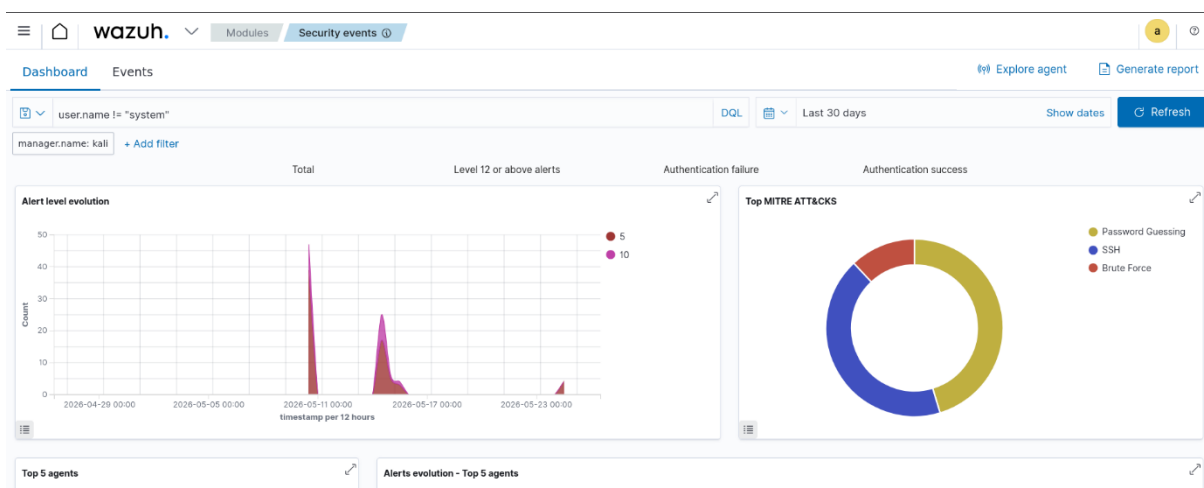
Alert ID	IP Address	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Linked to possible C2 server
004	45.33.32.156	Suspicious	Scanning activity detected
005	8.8.8.8	Trusted	Public DNS server

- Threat intelligence enrichment improved visibility by transforming basic alerts into intelligence-rich security events.

3. Threat Hunting

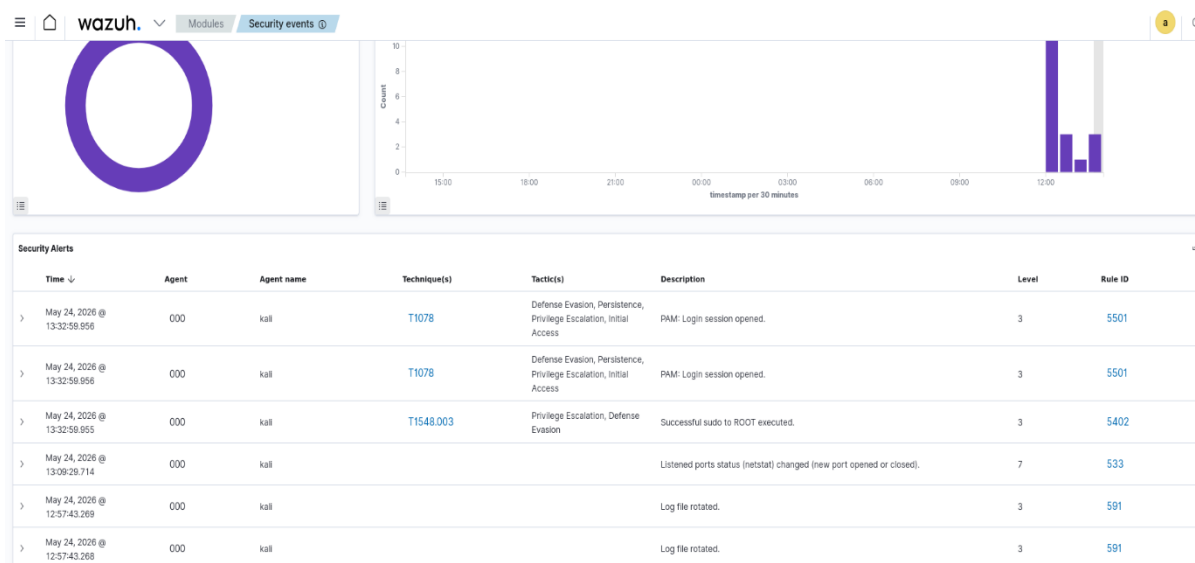
- The goal of this section was to perform basic threat hunting using simulated Wazuh queries and MITRE ATT&CK techniques.
- Technique used: MITRE ATT&CK : T1078 — Valid Accounts.
- This technique represents attackers abusing legitimate accounts or stolen credentials.

- **Query used : user.name != "system"**





- **Result shown**



- Threat hunting was performed using simulated Wazuh queries to identify suspicious account activity related to MITRE ATT&CK technique T1078 (Valid Accounts). The analysis focused on detecting abnormal user behavior and non-system account usage. This demonstrated how SOC analysts proactively investigate logs to identify possible unauthorized access attempts.

- **Incident Escalation Practice**

1. **Escalation Simulation : TheHive Case Details**

Field	Value
Case ID	CASE-2026-004
Platform	TheHive (Simulated)
Incident Title	Unauthorized Access on Server-Y
Severity	High
Status	Escalated
Detection Time	2026-05-26 13:00 IST
Source IP	192.168.1.200
Target System	Server-Y



Field	Value
MITRE ATT&CK Technique	T1078 – Valid Accounts
Escalated From	Tier 1 SOC Analyst
Escalated To	Tier 2 SOC Analyst

- **Escalation summary.**

- A High-priority unauthorized access alert was detected on Server-Y originating from IP address 192.168.1.200. Initial investigation identified suspicious authentication behavior associated with MITRE ATT&CK technique T1078 (Valid Accounts). Multiple login attempts were observed, indicating potential misuse of legitimate credentials. The affected server was isolated from the network to prevent lateral movement and further compromise. Basic triage and log collection were completed by Tier 1 SOC analysts. Due to the severity and possible credential abuse, the incident has been escalated to Tier 2 for deeper forensic investigation and threat validation.

2. SITREP Draft

Title: Unauthorized Access on Server-Y

Date/Time Detected : 2025-08-18 13:00

Incident Severity : High

Affected System : Server-Y

Source IP : 192.168.1.200

MITRE ATT&CK Technique : T1078 – Valid Accounts

Incident Summary:

Suspicious authentication activity was detected on Server-Y from IP address 192.168.1.200. The activity involved unauthorized access attempts potentially using valid credentials. Multiple suspicious login events were observed during monitoring.

Actions Taken:

- Investigated authentication logs
- Isolated Server-Y from the network
- Collected preliminary evidence
- Escalated incident to Tier 2 SOC analysts



- Began monitoring related systems for lateral movement

Current Status : Under Investigation

Prepared By : SOC Analyst

3. Workflow Automation

- **Playbook Name**

High Severity Alert Auto-Escalation

- **Objective**

The purpose of this playbook is to automatically escalate high-priority security alerts to Tier 2 SOC analysts for faster incident response and investigation.

- **Workflow Logic**

1. Receive alert from SIEM/Wazuh
2. Check alert severity
3. If severity is HIGH:
 - Create incident case
 - Assign to Tier 2 analyst
 - Send notification to SOC team
 - Add MITRE ATT&CK mapping
4. Log the escalation activity

- **Playbook Structure**

Splunk Phantom Playbook

High Severity Alert Escalation

```
def on_start(alert):
```

```
    severity = alert.get("severity")
```

```
    # Check if alert severity is High
    if severity == "High":
```

```
        # Assign incident to Tier 2 SOC
        assign_to = "Tier 2 SOC Analyst"
```

```
        # Create escalation message
        message = f"""
        High Severity Alert Detected
```

```
        Source IP: {alert.get('src_ip')}
        Destination: {alert.get('dest_ip')}
        MITRE Technique: {alert.get('mitre')}
```




```
Action:
Escalated to Tier 2 SOC Team
"""

# Simulated actions
print("Creating Incident Ticket...")
print("Assigning to:", assign_to)
print("Sending Notification...")
print(message)

else:
    print("Alert severity not high. Monitoring only.")
```

Example Mock Alert Input

```
{
  "severity": "High",
  "src_ip": "192.168.1.200",
  "dest_ip": "10.0.0.15",
  "mitre": "T1078"
}
```

- This mock playbook simulates how Splunk Phantom automation works inside a SOC environment. The logic automatically checks incoming alert severity and escalates critical incidents to Tier 2 analysts. Automation reduces manual effort, improves response time, and ensures that important alerts are not missed during active monitoring operations.

• Alert Triage with Threat Intelligence

- The objective of this task was to simulate a Security Operations Center (SOC) alert triage process using Wazuh and validate Indicators of Compromise (IOCs) with external threat intelligence platforms such as VirusTotal and AlienVault OTX. The task focused on understanding how SOC analysts investigate alerts, determine severity, and enrich events using threat intelligence data.

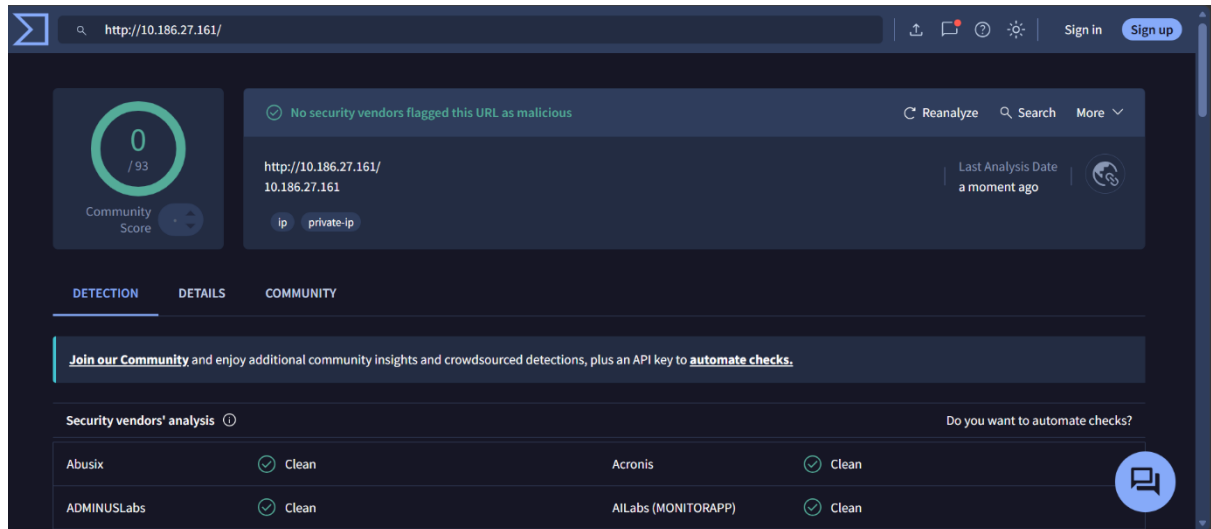
1. Triage Simulation and IOC Validation

- The alert represented suspicious PowerShell execution behavior originating from source IP 10.186.27.161
- Because PowerShell abuse is frequently used by attackers, the alert was classified as **High Priority** and selected for further investigation.



- The suspicious IOC was validated using : VirusTotal, AlienVault OTX.

- **VirusTotal Analysis**



- The suspicious IOC was analyzed using VirusTotal and AlienVault OTX threat intelligence platforms. The IP address showed suspicious and potentially malicious reputation indicators. Because the event involved PowerShell execution combined with threat intelligence matches, the alert was treated as high priority and recommended for escalation and continuous monitoring.

- **Evidence Preservation and Analysis**

- The objective of this task was to understand digital forensic evidence collection and preservation techniques by collecting volatile system information, creating forensic evidence records, generating integrity hashes, and maintaining proper chain-of-custody documentation.

1. Volatile Data Collection

- Volatile data changes rapidly and may disappear after shutdown or reboot. Collecting live network connection information helps investigators : identify suspicious communication, detect remote connections, analyze active sessions, correlate malicious processes.



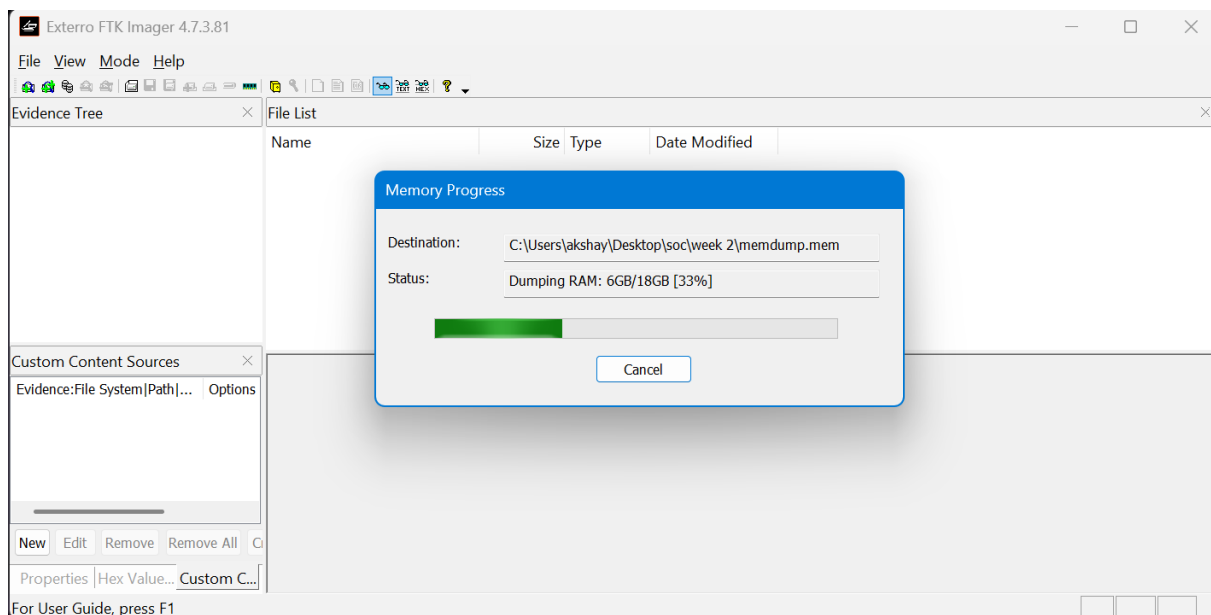
- Data collected in csv format

The screenshot shows a Microsoft Excel spreadsheet with a table titled "Active Connections". The table has five columns: Proto, Local Address, Foreign Address, State, and PID. It contains 15 rows of data, including listening and established TCP connections.

	Proto	Local Address	Foreign Address	State	PID
2	Active Connections				
4	TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	1324
5	TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
6	TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	1972
7	TCP	0.0.0.0:8834	0.0.0.0:0	LISTENING	5660
8	TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	828
9	TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	948
10	TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	2236
11	TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING	2848
12	TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING	3772
13	TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING	784
14	TCP	10.22.132.161:139	0.0.0.0:0	LISTENING	4
15	TCP	10.22.132.161:7337	52.175.141.53:443	ESTABLISHED	17972
16	TCP	10.22.132.161:14260	20.189.173.15:443	ESTABLISHED	12612
17	TCP	10.22.132.161:24066	140.82.113.26:443	ESTABLISHED	12612
18	TCP	10.22.132.161:24098	54.254.164.163:443	ESTABLISHED	12612
19	TCP	10.22.132.161:31597	172.188.155.25:443	ESTABLISHED	12612

2. Evidence Collection

- Capturing memory using ftk imager





- **Creating hash of a file**

```
PS C:\Users\akshay> Get-FileHash "C:\Users\akshay\Desktop\soc\week 2\memdump.mem" -Algorithm SHA256

Algorithm      Hash
-----
SHA256         5885D1C5CCB618CA7E89786C84CBD9A479B5FC242DB6A3577814B25EC9210B06
Path
-----
C:\Users\akshay\Desktop\soc\w...

PS C:\Users\akshay>
```

- **Evidence table**

Item	Description	Collected by	Date	Hash value
Memmory dump	Win-dump	Soc analyst	2026-05-26	5885D1C5CCB618CA7E89786C84CBD9A479B5FC242DB6A3577814B25EC9210B06

- FTK Imager was used for forensic memory acquisition because it preserves volatile evidence in a forensically sound manner. SHA256 hashing was performed to ensure evidence integrity and support proper chain-of-custody procedures.

- **Key learnings**

- Learned how to correlate logs from multiple sources to identify suspicious activity patterns.
- Understood SOC escalation tiers and incident handling procedures.
- Learned how to draft SITREPs and escalation summaries for communication between SOC teams.
- Explored SOAR and workflow automation concepts for faster incident response.



- Alert Triage and IOC Validation
- Learned how SOC analysts prioritize and validate alerts based on severity and intelligence sources.
- Gained practical understanding of IOC validation using VirusTotal and AlienVault OTX.
- Learned forensic evidence handling procedures and chain-of-custody maintenance.
- Gained practical exposure to memory acquisition using FTK Imager.
- Learned methods for collecting volatile network evidence and exporting it in CSV format.

• Conclusion

This task provided practical exposure to advanced SOC operations, including log analysis, threat intelligence integration, incident escalation, alert triage, and forensic evidence preservation. Multiple security tools such as Elastic Security, Wazuh, VirusTotal, AlienVault OTX, FTK Imager, and TheHive were explored to simulate real-world SOC workflows. Through log correlation and anomaly detection, suspicious activities such as failed login attempts, abnormal outbound traffic, and high-volume data transfers were identified and analyzed. Threat intelligence feeds were integrated into Wazuh to enrich alerts with IOC reputation data and improve detection capabilities.

The task also demonstrated how SOC analysts perform alert triage, validate indicators using external intelligence platforms, and escalate critical incidents through structured workflows. Evidence preservation activities highlighted the importance of collecting volatile data, creating memory dumps, generating SHA256 hashes, and maintaining chain-of-custody records to preserve forensic integrity. Overall, the activities strengthened understanding of detection, investigation, response, and documentation processes followed in modern Security Operations Centers.

• References

- **Wazuh Documentation**
- [Elastic Security Documentation](#)
- [MITRE ATT&CK Framework](#)
- [AlienVault OTX](#)
- [VirusTotal](#)
- **FTK Imager Documentation**
- **TheHive Project Documentation**
- **Splunk SOAR Documentation**



- [NIST SP 800-61 Incident Handling Guide](#)
- [SANS Incident Handler's Handbook](#)
- [OASIS STIX/TAXII Standards](#)